

PROJECT DOCUMENTATION

Smart Visitor Access Management System

ABSTRACT

The Smart Visitor Access Management System is an enterprise-level ServiceNow application developed to automate and streamline the visitor management process within an organization. The system provides a centralized platform for visitor registration, approval management, security validation, visitor tracking, and reporting.

The application eliminates manual visitor handling by automating request submission, approval workflows, email notifications, visitor badge generation, check-in/check-out operations, and VIP visitor processing. Advanced features such as blacklist verification, duplicate visitor detection, automated expiry management, and dashboard analytics improve organizational security and operational efficiency.

The system ensures that only authorized visitors are granted access while providing complete visibility into visitor activities throughout their lifecycle.

INTRODUCTION

Visitor management plays a critical role in maintaining organizational security, operational control, and compliance. Organizations regularly receive visitors such as clients, vendors, auditors, consultants, maintenance personnel, and interview candidates.

Traditional visitor management methods often depend on manual registers, spreadsheets, emails, and phone-based approvals. These approaches result in delays, data inconsistencies, lack of visibility, and security concerns.

To overcome these limitations, the Smart Visitor Access Management System was developed using the ServiceNow platform. The solution automates the complete visitor lifecycle from registration to exit tracking while ensuring security, transparency, and accountability.

PROBLEM STATEMENT

Many organizations continue to manage visitor activities manually, resulting in several operational and security challenges.

Existing Challenges

- Manual visitor registration process
- Delayed approval mechanisms
- Poor visitor tracking capabilities
- Security vulnerabilities
- Lack of centralized records

- Duplicate visitor registrations
- Inefficient communication
- Limited reporting and analytics
- Increased administrative workload

These issues negatively impact productivity, security, and visitor experience.

PROJECT OBJECTIVES

Primary Objectives

- Automate visitor registration.
- Streamline approval workflows.
- Improve visitor security management.
- Enable real-time visitor tracking.
- Automate notification processes.
- Maintain complete visitor history.
- Reduce manual intervention.

Secondary Objectives

- Improve operational efficiency.
- Enhance user experience.
- Increase process transparency.
- Generate analytical reports.
- Support VIP visitor management.
- Strengthen organizational security.

SYSTEM OVERVIEW

The Smart Visitor Access Management System automates the complete visitor lifecycle using ServiceNow technologies.

Core Modules

Visitor Registration Module

Captures visitor information including personal details, visit purpose, visit date, and host employee information.

Validation Module

Performs phone number validation, date validation, duplicate detection, and blacklist verification.

Approval Management Module

Routes visitor requests through an automated approval workflow.

Notification Module

Sends automated notifications to security personnel and stakeholders.

Security Operations Module

Manages visitor entry and exit activities.

Dashboard Module

Provides real-time analytics and monitoring capabilities.

PLATFORM AND TECHNOLOGIES USED

Platform

ServiceNow Developer Instance

Technologies and Components

- ServiceNow Studio
- Custom Tables
- Form Designer
- Client Scripts
- Business Rules
- Flow Designer
- Email Notifications
- UI Actions
- Scheduled Jobs
- Dashboards & Reports
- Access Controls

APPLICATION DEVELOPMENT

Application Name

Smart Visitor Access Management System

The application was developed using ServiceNow Studio as a custom scoped application.

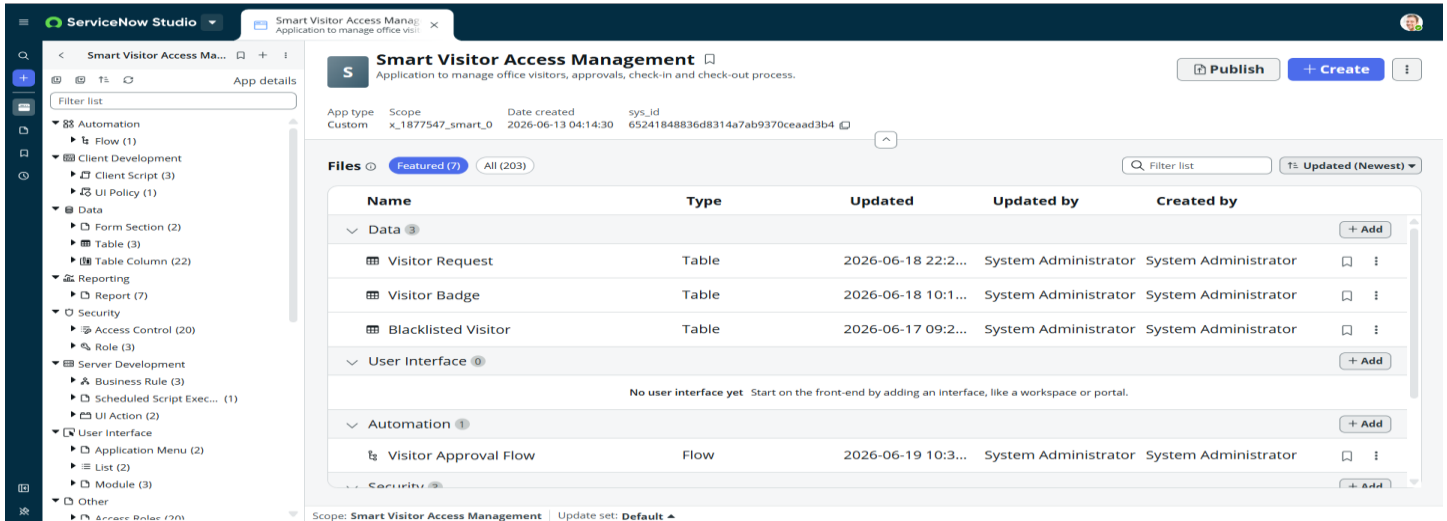
Purpose

To provide a centralized and automated visitor management solution.

Application Components

- Visitor Request Table

- Visitor Badge Table
- Blacklisted Visitor Table
- Visitor Approval Flow
- Email Notifications
- Security Operations
- Dashboard Reports



DATABASE DESIGN

Table Name

Visitor Request

The Visitor Request table serves as the primary repository for storing visitor information.

Fields Implemented

Field Name	Description
Visitor ID	Unique Visitor Identifier
Visitor Name	Visitor Full Name
Phone Number	Contact Number
Email	Visitor Email Address
Company Name	Visitor Organization
Purpose	Visit Purpose
Visit Date	Scheduled Visit Date
Host Employee	Employee Being Visited

Status	Current Request Status
VIP Visitor	VIP Indicator
Priority	Priority Level
Check-In Time	Visitor Entry Time
Check-Out Time	Visitor Exit Time

Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)	32		false
Created	Date/Time	(empty)	40		false
Priority	Choice	(empty)	40		false
Email	String	(empty)	100		false
Visitor ID	String	(empty)	40		false
Check In Time	Date/Time	(empty)	40		false
Badge Number	String	(empty)	40		false
Status	Choice	(empty)	40		false
Updated by	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Company Name	String	(empty)	40		false
Host Employee	Reference	User	32		false
Purpose	Choice	(empty)	40		false
Check Out Time	Date/Time	(empty)	40		false
Number	String	(empty)	40	javascript:global.getNextObjNumberPadded();	false
Updated	Date/Time	(empty)	40		false
Visitor Name	String	(empty)	100		false
Visit Date	Date	(empty)	40		false
Created by	String	(empty)	40		false
VIP Visitor	True/False	(empty)	40		false

FORM DEVELOPMENT

A structured visitor registration form was developed to collect visitor information.

Visitor Details

- Visitor Name
- Phone Number
- Email Address
- Company Name

Visit Details

- Purpose
- Visit Date
- Host Employee
- VIP Visitor

Benefits

- Standardized data collection
- Improved usability
- Reduced errors
- Better user experience

Visitor Request - Create VIS0096

Application scope: Smart Visitor A...
Update set: Default [Smart Visitor...]

Visitor Request
New record

Submit

VALIDATION IMPLEMENTATION

Phone Number Validation

A Client Script validates visitor phone numbers.

Features

- Accepts only 10-digit numbers
- Prevents invalid entries
- Displays validation messages

Visit Date Validation

The system prevents selection of past dates.

Features

- Accepts only current and future dates
- Ensures valid scheduling

Duplicate Visitor Detection

A Business Rule prevents duplicate visitor registrations.

Logic

Phone Number + Visit Date

Benefits

- Eliminates duplicate records
- Improves data quality

Blacklisted Visitor Validation

A Business Rule verifies visitor information against blacklist records.

Benefits

- Prevents unauthorized access
- Enhances security controls

WORKFLOW AUTOMATION

Flow Name

Visitor Approval Flow

The workflow was developed using ServiceNow Flow Designer.

Trigger

Visitor Request Created

Approval Action

Ask For Approval

Conditional Logic

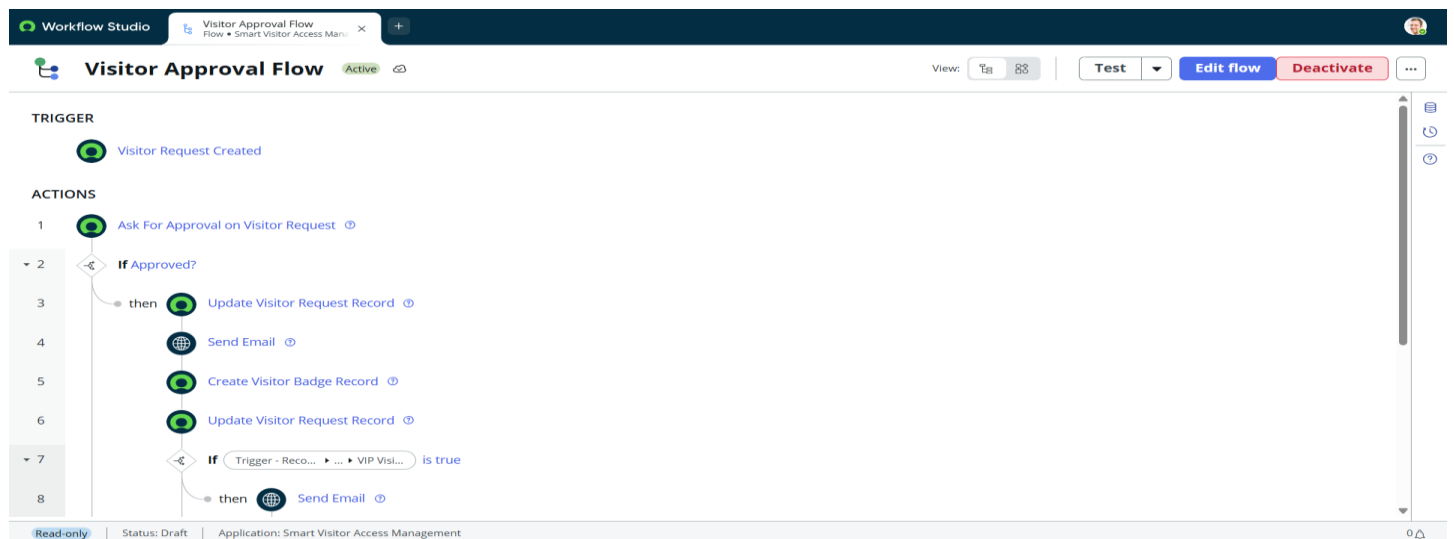
Approval State = Approved

Automated Activities

- Update Visitor Record
- Send Email Notification
- Create Visitor Badge Record
- VIP Visitor Processing
- Security Notification

Benefits

- Automated approval process
- Reduced manual intervention
- Faster request processing



VISITOR BADGE MANAGEMENT

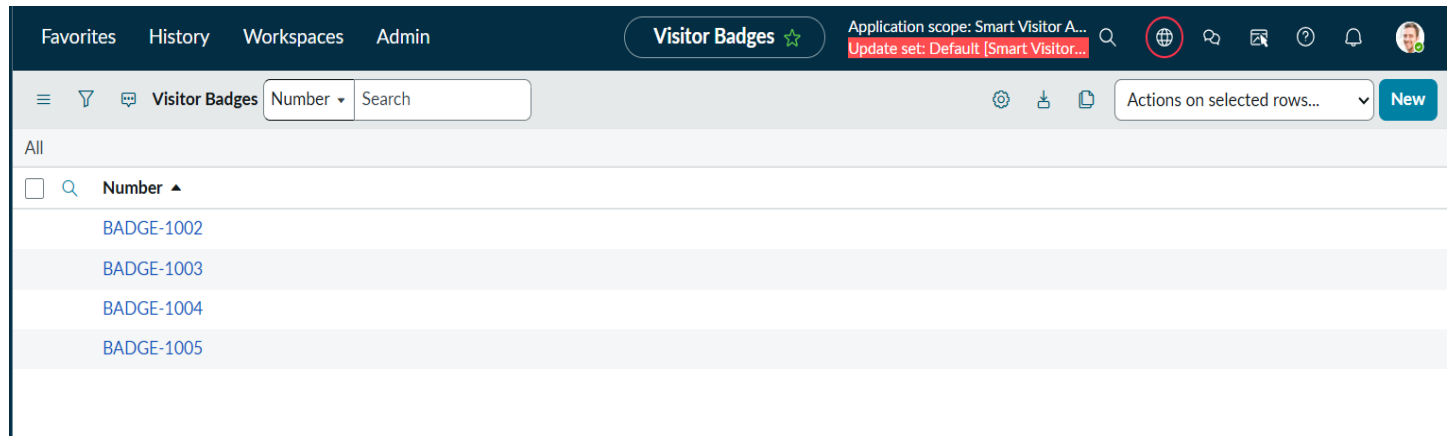
The system automatically creates visitor badge records after approval.

Features

- Unique badge generation
- Visitor identification
- Security support

Benefits

- Improved access control
- Faster visitor verification



SECURITY OPERATIONS

Visitor Check-In

The Check-In functionality records visitor entry details.

Features

- Entry time capture
- Status update
- Visitor tracking

Visitor Check-Out

The Check-Out functionality records visitor exit details.

Features

- Exit time capture
- Status update
- Audit tracking

VIP VISITOR MANAGEMENT

A dedicated VIP handling mechanism was implemented.

Features

- Automatic High Priority Assignment

- VIP Notifications
- Enhanced Monitoring

Benefits

- Better visitor experience
- Improved security preparedness

AUTOMATED EXPIRY MANAGEMENT

A Scheduled Job automatically expires outdated visitor requests.

Functionality

- Daily execution
- Automatic status updates
- Record maintenance

Benefits

- Improved database quality
- Reduced manual maintenance

DASHBOARD AND REPORTING

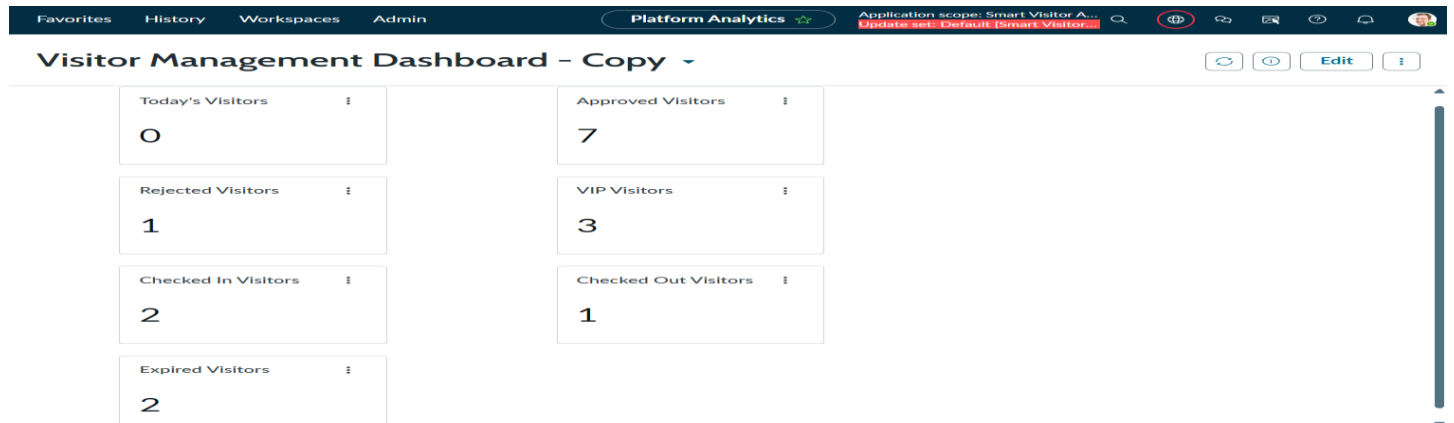
A comprehensive Visitor Management Dashboard was developed.

Reports Implemented

- Today's Visitors
- Approved Visitors
- Rejected Visitors
- Checked-In Visitors
- Checked-Out Visitors
- VIP Visitors
- Expired Visitors

Dashboard Benefits

- Real-time monitoring
- Better decision making
- Operational visibility
- Trend analysis



RESULTS AND ACHIEVEMENTS

Project Achievements

- ✓ Automated Visitor Registration
- ✓ Automated Approval Workflow
- ✓ Automated Notifications
- ✓ Visitor Badge Generation
- ✓ Blacklist Verification
- ✓ VIP Visitor Handling
- ✓ Check-In / Check-Out Tracking
- ✓ Dashboard Analytics

Business Benefits

- ✓ Improved Security
- ✓ Reduced Processing Time
- ✓ Increased Transparency
- ✓ Better Visitor Tracking
- ✓ Enhanced Operational Efficiency
- ✓ Reduced Administrative Workload

CONCLUSION

The Smart Visitor Access Management System successfully automates and digitizes the complete visitor management lifecycle using the ServiceNow platform. The application integrates visitor registration, approval automation, security validation, visitor badge generation, VIP management, check-in/check-out tracking, automated notifications, and dashboard analytics into a unified solution.

The project demonstrates how ServiceNow can be leveraged to build secure, scalable, and enterprise-grade business applications that improve operational efficiency, strengthen security, and enhance user experience.